

Parameters, Weights, and Biases

What Are Parameters?

Parameters are the learnable numbers inside a neural network. During training, the model adjusts these numbers millions of times until they encode patterns from the training data. When people say 'GPT-3 has 175 billion parameters', they mean 175 billion individual numbers that store the model's 'knowledge'.

An LLM's 'knowledge' lives entirely in its parameters — not in a fact database you can directly query.

Weights vs. Biases

Weights (W): Numbers that scale the strength of connections between neurons. They determine how much influence one neuron has on another.

Biases (b): A constant added to the weighted sum before the activation function. They shift the output, allowing the model to fit data that doesn't pass through the origin.

The computation in each neuron is: $\text{output} = \text{activation}(W \cdot \text{input} + b)$

Analogy: weights are like the volume knobs on a mixing board; biases are like a baseline volume applied before the knobs.

What Does 'Large' in LLM Mean?

'Large' refers to the number of parameters, not to intelligence or capability directly:

Model	Parameters (approx.)
GPT-2 (2019)	1.5 billion
GPT-3 (2020)	175 billion
PaLM (2022)	540 billion
GPT-4 (2023)	~1 trillion (est.)

Parameters ≠ a Fact Database

A common misconception is that a model 'looks up' facts stored in a table. In reality, facts are distributed across billions of parameter values — the model reconstructs them by computing through its layers. This is why LLMs can 'hallucinate': they generate plausible parameter-based reconstructions that are not always accurate.

Key Takeaways

- Parameters are all the learnable numbers that define a model's behaviour.
- Weights scale inter-neuron connections; biases shift outputs.
- 'Large' in LLM refers to parameter count (billions to trillions).
- Knowledge is distributed across parameters — not stored in a searchable database.

Quick-Revision Questions

- What is the difference between a weight and a bias?
- Why does a large parameter count not guarantee factual accuracy?
- Write the computation for a single neuron's output.